Bishnu Ramachandran

Summary

M.A.Sc GRA in Materials Engineering, with a dual degree in Mathematics and Chemical Engineering, and first-author publication on ML applications in polymer science (Soft Matter, RSC). Experienced in interdisciplinary research, machine learning, data analysis, and analytics consulting. Skills spanning Python, process simulation, SQL, ML pipelines, and interpretable AI, with experience communicating technical results and mentoring junior colleagues.

Education

University of British Columbia (UBC) — Vancouver, Canada
M. A. Sc in Materials Engineering — Aug'26 – Ongoing
Graduate Research Assistant (Supervisor: Karthik Akkiraju)

Birla Institute of Science and Technology (BITS) Pilani — Hyderabad, India
Integrated Dual Degree, MSc Mathematics & B.E Chemical Engineering — Sep'20 – Jun'25
CGPA: 8.48

Research Experience

University of British Columbia (UBC) — Vancouver, Canada
Data-Driven Modelling in Absolute Sustainability — Aug'26 - Ongoing

- Multi-scale modelling utilizing quantum ML tools, process flows, and system dynamics
- Designing robust and explainable decision frameworks for sustainability assessments
- Evaluation of planetary level metrics for process systems alongside human QoL impact

Birla Institute of Science and Technology (BITS) Pilani — Hyderabad, India
Machine Learning in Polymer Science — Sep'23 – May'24

- Analyzed the complex physics of polymer systems with critical industrial applications
- Evaluated multiple models on imbalanced datasets and improved model accuracy by 30%
- Leveraged interpretable ML (SHAP) for reducing empirical resource expenditure
- Engineered a free-to-access web application to democratize the model's accessibility

Publications

1. Morphology prediction for polymer blend thin films using machine learning

Journal : Soft Matter (Royal Society of Chemistry)
Publication Date : 03 Jun 2025
Citation : Soft Matter, 2025,21, 5284-5295

2. On A Fourth Order Boundary Value Problem Arising In Elastic Beam Analysis

Journal : Applied Mathematics E-Notes
Publication Date : 16 Nov 2025
Citation : Applied Mathematics E-Notes, 2025, 25, 441-454

Professional Experience

Indus Insights Gurgaon, Haryana
Junior Associate Jun'24 - Jun'25
Associate II Jul'25 - Sep'25
Senior Associate Oct'25 - Jul'26

- Reporting to multiple executive stakeholders for a US based financial solutions provider
- Leveraged Python, VBA and SQL for automation leading to a 30% reduction in SLA
- Conducted non-spread waiver segment analysis, re-capturing ~ $ 500k of annual income
- Built dynamic roll rates-based yield forecasting models reducing portfolio losses by 15%
- Designed an automated Python pipeline for sharing weekly KPIs to stakeholders 80% faster
- Mentored junior analysts on technical tools and stakeholder management

Skills

Programming:

- Python, MATLAB, SQL, Java, HYSYS
- SSMS, Spring Boot, Amazon RedShift

Machine Learning and Data Science:

- Scikit-Learn, SHAP, joblib, Classification and Regression Models
- Interpretable ML, Optimization, Neural Networks, Time Series Modelling

Data and Visualization Tools:

- Excel, PowerPoint, SPSS Modeler, Tableau, SSRS, Power BI

Soft Skills:

- Client management, deliverable ownership, bandwidth assessment
- Agile / sprint based workflows, independent problem-solving, cross-collaborative performance